

## Technical Datasheet 4G Gateway

### Product Overview:-

The 4G Gateway is an advanced networking device designed to provide seamless connectivity and data management capabilities for various applications. With support for 4G LTE connectivity, cloud connectivity via MQTT protocol, and robust storage options, it offers a comprehensive solution for remote monitoring and control systems. Equipped with Modbus RS485 port and micro Sim compatibility, it caters to a wide range of industrial and IoT applications. The intuitive web-based UI interface allows for easy configuration and management, making it an ideal choice for both professionals and enthusiasts.



## Specifications

### General:-

- **4G LTE Connectivity:** Utilizes 4G LTE technology for reliable and high-speed internet access. If 4G is not available, the system automatically switches to Wi-Fi. If both 4G and Wi-Fi are unavailable, the data is securely stored in the SD card for later upload.
- **Cloud Connectivity (MQTT):** Seamlessly connects to cloud platforms using the MQTT protocol for efficient data transmission and management.
- **Event Storage:** Offers on board event storage, capable of storing up to 50,000 events or more. Includes an external memory card slot for easy data retrieval in the event of 4G network issues.
- **Modbus RS485 Port:** Features 1 port of Modbus RS485 for integration with industrial control systems and sensors.
- **GPS Location Data:** Provides GPS location data for monitoring of remote equipment.
- **Micro Sim Compatibility:** Supports micro-Sim card for flexible and convenient connectivity option.
- **Web-based UI Interface:** Intuitive web-based user interface for easy configuration, monitoring, and management.
- **Dimensions:** 110 mm L x 110 mm B x 50 mm H.
- **Power:** Input Power – 15-40 VDC, Typical – 12V DC @ 150mA.
- **Operating Temperature:** 0 – 60C (32 ~ 140F).
- **Storage Temperature:** 20 - 70C (-4 ~ 158F).
- **Storage Humidity:** 5 ~ 95 % RH, non – Condensing.

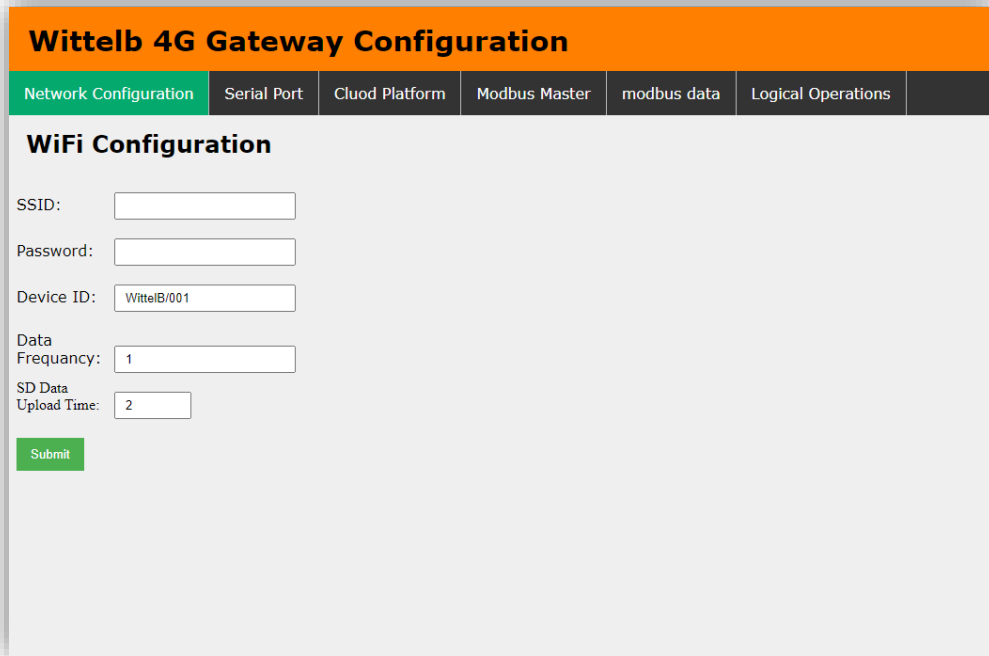
### LED Indicators:-

- **POWER:** Device power status.
- **NETWORK:** Indicates 4G/Wi-Fi network connectivity.
- **CLOUD:** Shows MQTT cloud connection status.
- **POLL:** POLL/WI-FI: Indicates Modbus polling activity / Wi-Fi mode
- **RX:** Data received from RS485.
- **TX:** Data transmitted to RS485.



### Configuration Settings:

- Power up the module by connecting a 12 V DC supply.
- Insert the SIM card into the designated slot.
- Place the Micro SD card into its designated slot.
- Connect MODBUS RTU Multiple Slave devices (up to 31) to the A & B Terminal Of Module.
- Access the module's configuration hotspot (ID: WIN-4G-GATEWAY, PASS: 123456789).
- Default IP Address of Module is "192.168.1.100/cpp1".
- Upon entering the provided IP address, a configuration webpage Pages will be accessible.

**Page 1: -**

- Enter Device ID (Should be unique for every device).
- Enter data frequency in min to receive data on cloud.
- Enter time to upload SD card data to cloud.

**Page 2:-**

On second page you can set device Baud Rate (9600 / 19200 / 38400 / 57600 / 115200), Data Bits (5 / 6 / 7 / 8), Parity (None / Odd / Even), and Stop Bits (1 / 2) for RS485 Communication

**Wittelb 4G Gateway Configuration**

Network Configuration

Serial Port

Cloud Platform

Modbus Master

modbus data

Logical Operations

**Rs485 Configuration**

Baud Rate: 9600

Data bits: 8

Parity: None

Stop Bits: 1

Save RS485 Configuration

### Page 3:-

- Navigate to page 3 on the cloud platform.
- Input the Host IP or Domain.
- Enter the port Number.
- Enter the topics for publishing and receiving messages.
- Provide a username and password if required.
- Allow a few seconds for the module to restart and establish a connection to MQTT.
- Select Wifi and 4G from Which Source you need to send data over MQTT

**Wittelb 4G Gateway Configuration**

Network Configuration | Serial Port | **Cloud Platform** | Modbus Master | Monitor | About

### Cloud Platform Configuration

**Network Mode**

WiFi ☒ ☐   
Current Mode: 4G Network

Host IP or Domain:

Port:

Publish Topic:

Publish Topic (SD Card):

Subscribe Topic:

User Name:

Password:

**Save Configuration**

## Page 4:-

- Go to page 4 on the Modbus Master interface.
- Input the name and slave address for the designated slave.
- Specify the register address from which you want to read slave data.
- You can enter only the Register which you need to read.
- Select the Data Type: UINT16, Float32 Big Endian, Float32 Little Endian, Float32 BE Byte Swap, or Float32 LE Byte Swap.
- Enter the function code corresponding to the desired slave.

**Wittelb 4G Gateway Configuration**

Network Configuration | Serial Port | Cloud Platform | **Modbus Master** | Monitor | About

### Modbus Master Configuration

Saved successfully (2 rows)

| Name | Slave Address | Register | Function Code | Data Type             | Actions |
|------|---------------|----------|---------------|-----------------------|---------|
| One  | 1             | 3        | 3             | Float32 Little Endian | Delete  |
| Two  | 1             | 5        | 3             | Float32 Little Endian | Delete  |

**Add Row** **Submit**

## Page5:-

- On page 5 (Monitor), view live Modbus register values for all configured entries.
- Table auto-refreshes every 7 seconds showing: Slave Address, Register, Function Code, Tag Name, Data Type, Live Value, and Status (OK / FAIL)
- Use Slave Address and Register filter fields to narrow down the live view.

**Wittelb 4G Gateway Configuration**

Network Configuration
 Serial Port
 Cloud Platform
 Modbus Master
 **Monitor**
 About

### Modbus Live Monitor

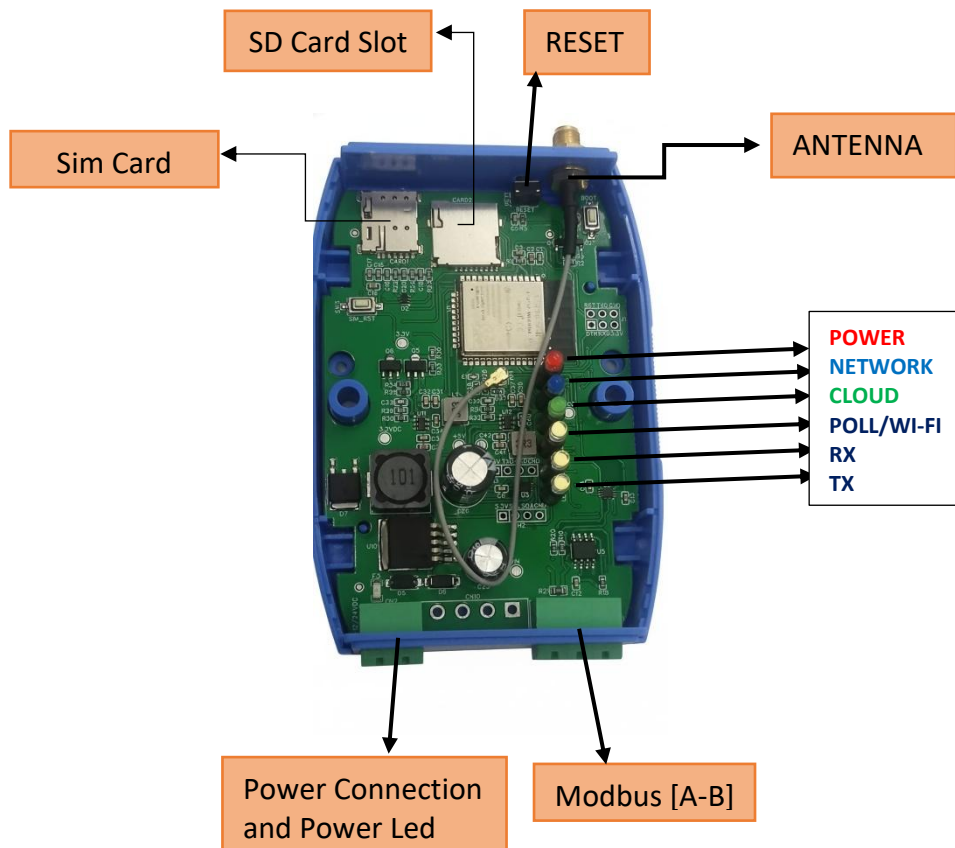
Total Registers: **2**
 Slaves Active: **1**
 Showing: **2**

Slave Address: 
 Register: 
Filter
Clear

Live — 2 registers | Updated: 17:32:11
 Next refresh in 5s

| Sr.No | Slave Address | Register | Function Code | Name / Tag | Data Type   | Live Value | Status | Last Updated |
|-------|---------------|----------|---------------|------------|-------------|------------|--------|--------------|
| 1     | 1             | 3        | FC3           | One        | FLOAT BA-DC | ERR        | X FAIL | —            |
| 2     | 1             | 5        | FC3           | Two        | FLOAT BA-DC | ERR        | X FAIL | —            |

### 4G Gateway connection Diagram: -



### Additional Features: -

Communication ports are isolated

Input power reverse polarity safety

ESD Safety IEC 61000-4-2,  $\pm 30\text{KV}$  contact,  $\pm 30\text{KV}$  air

EFT IEC 61000-4-4, 50A (5/50ms) 400V isolation.

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